

Metaphysical Foundations, Channel Triangulation, and Functorial Analogy Transfer

System Architecture Invariant Manifest

September 28, 2026

1 Introduction and Ontological Modalities

Following the formalisms of Edward Zalta’s Abstract Object Theory and Situation Theory, we isolate three distinct modal world-lines within a single unified processing manifold:

1. **The Historical Immutable World:** Rigidly designated actualities (s_{actual}) frozen in the past (e.g., Moses, David, Paul).
2. **The Fictional Mutable World:** Abstract objects encoding properties inside an isolated situation domain (s'), completely open to counterfactual planning rewrites (e.g., John, Judith, Pierre).
3. **The Indexical Real World:** The concrete layer where sovereign actions occur in real time.

2 The DSL Discretization Interface and Continuous Superposition

Continuous environments present an infinite, unsearchable behavior space. We address this challenge by introducing a LISP-like Domain-Specific Language (DSL) that functions as a structural discretization filter.

2.1 The Accuracy Advantage of Continuous Superposition

In a traditional discrete tree, a model must commit prematurely to either a True or False branch. Near precarious physical boundaries, this introduces a truncation error that can cause down-stream instabilities. By maintaining

alternative trajectories in a continuous superposition field, multiple overlapping constraints are evaluated simultaneously. Stable paths experience constructive interference, while high-error paths undergo destructive phase-cancellation, mirroring the active wave-reflection compensation scripts.

3 The Principle of Triangulation via Channel Infomorphisms

While Jon Barwise formulated Channel Composition to track how information commutes through distributed networks, we deploy the Principle of Triangulation to secure cross-model analogies. If three completely independent structural representations (FCA Parthood, Graph Topologies, and Fallacy Patterns) all arrive at the exact same leaf category, the analogy is mathematically finalized.

4 Functorial Cross-Domain Analogy Transfer (Tree #11)

When a structural similarity is inferred across two completely distinct physical universes (e.g., Thermodynamic Wave Reflections and Drone Aerodynamic Wind-Gusts), the relationship maps to a Category Theory Functor ($F : \mathcal{C} \rightarrow \mathcal{D}$).

Instead of searching for raw textual keywords, Tree #11 checks for abstract symbolic invariants: `<group_type: semi_simple_continuous, disruption: active_reflections>`. When both systems hit the identical terminal leaf node (`lie_algebra_phase_cancellation`), the DSL interpreter dynamically reuses the exact same underlying control law expression program (`mul - 1`), transferring problem-solving experience across separate physical manifestations.